# Indirect Self-tuning Regulators (Åström & Wittenmark, Examples 3.4–3.6)

Adaptive Control, Homework 3

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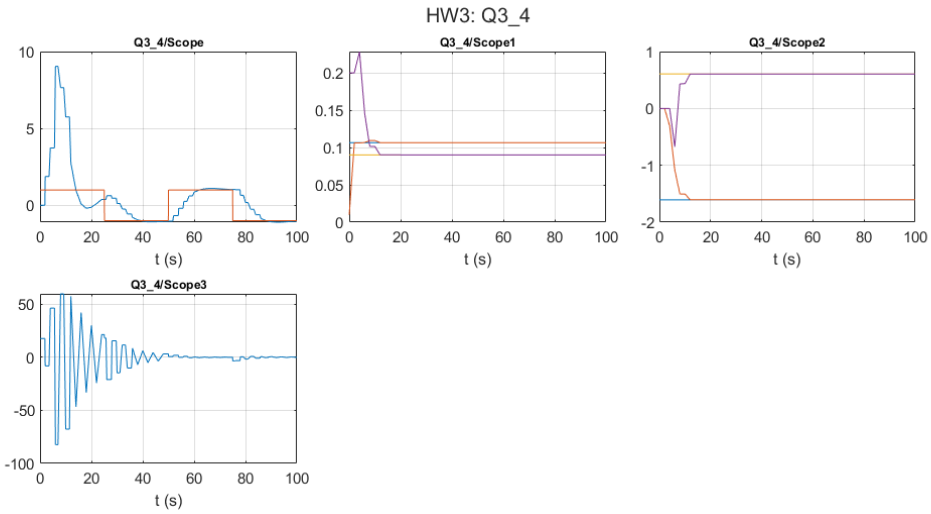
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All three cases are subsystems of `HW3.slx`. Each one combines RLS estimation of the plant with a pole-placement design recomputed at every step from the current estimates. For each case my report puts my simulation next to the book's figure: output tracking of a square-wave reference, the control signal, and the parameter estimates.

**Example 3.4: with zero cancellation.** The controller cancels the process zero.

**Example 3.5: without zero cancellation.** The process zero is kept in the closed-loop response.

**Example 3.6: continuous-time STR.** The same design applied to a continuous-time plant model.




Re-run in R2023b: example 3.4, output and reference, estimates and control signal.